

Exercise Set 4 - Part 2

Problem 4

Pfeifer/Stephan (2019) investigate whether there are gender differences in the fairness perception of one's own wage. This is particularly relevant with respect to negotiations about subsequent wage growth. Table 2 shows average marginal effects for the probability to perceive the own wage as fair resulting from random effects binary probit regressions.

Table 2: Average marginal effects from probits for the probability to perceive the own wage as fair

	(1)	(2)
Female	0.0454 (0.0126)	
Female $\times$ single		0.0263 (0.0200)
Female $\times$ married		0.0498 (0.0156)
Hourly net wage in Euros	0.0353 (0.0015)	YES
Actual weekly working hours	-0.0062 (0.0005)	YES
Marital status		YES
Number of observations	26,051	23,216
Number of individuals	12,213	10,967

Note: Standard errors are in parentheses. Sample includes only singles and married (not separated) persons.

Source: Pfeifer, C., Stephan, G. (2019). Why women do not ask: gender differences in fairness perceptions of own wages and subsequent wage growth. Cambridge Journal of Economics, 43:295–310.

- 4.1 Name a reason why the authors prefer a random effects model over a fixed effects model?
- 4.2 Interpret the estimation results for the three main explanatory variables in specification (1).
- 4.3 Specification (2) reports interaction terms of female with being single or married. Is there a difference between married and single women?

Problem 5

Lemieux et al. (2009) analyse wages in performance-pay jobs. Using panel data for workers in the US, they estimate OLS and fixed effects regressions of log wages on a dummy for whether or not the worker receives performance-pay (e.g. bonuses, stock options, commissions). The model also includes experience (number of years in the workforce, measured in 20 years) and tenure (number of years at this specific job, measured in 10 years) as well as further control variables. Table 3 shows the estimation results for the coefficients. Standard errors are in parentheses.

Table 3: Skills-related wage differentials and performance-pay (PP) Jobs

	(1) OLS	(2) FE
Performance-pay job dummy	-0.4526 (0.1019)	-0.2061 (0.0723)
Education (measured in years)	0.0637 (0.0040)	0.0167 (0.0091)
Education $\times$ performance-pay job	0.0365 (0.0071)	0.0169 (0.0048)
Experience (measured in 20 years)	0.3010 (0.0294)	0.4545 (0.1258)
Experience $\times$ performance-pay job	0.1162 (0.0584)	0.0149 (0.0501)
Tenure (measured in 10 years)	0.2262 (0.0154)	0.1158 (0.0129)
Tenure $\times$ performance-pay jobs	-0.0666 (0.0301)	0.0278 (0.0237)

Note: Column (1) shows results from an OLS regression. Column (2) shows results from a fixed effects regression (worker fixed effect). Dependent variable is log wage. Standard errors are in parentheses.

Source: Lemieux, T., Macleod, W. B., and Parent, D. (2009). Performance pay and wage inequality. *The Quarterly Journal of Economics*, 124(1):1–49.

- 5.1 Consider the results in column (1). Are the returns to education higher for performance-pay jobs than for non-performance-pay jobs? Provide and interpret the difference in returns. Assess whether the difference is statistically significant.
- 5.2 Consider the results in Table 3, column (1). What is the effect of performance-pay for a worker with 16 years of education, 20 years of experience, and 10 years of tenure? Show your calculations and give an interpretation.
- 5.3 The estimate of years of education falls from 0.0637 in column (1) to 0.0167 in column (2). Provide an explanation for this finding. Assume that ability is an unobserved variable.

Problem 6

Analyse the demand for health care using the data set by Riphahn et al. (2003).

- 6.1 Download the data set from the *Journal of Applied Econometrics Data Archive*. Load the data set into R and familiarise yourself with the data set. Rename the variables according to the readme-file, set the data as panel data and report the number of observations.
- 6.2 Dichotomise the variable *docvis*, taking the value 1 if a person has been to the doctor in the last three months and 0 otherwise.
- 6.3 Keep the variables needed for your analysis. Present descriptive statistics for these variables. Compare the descriptive statistics of people who have been to the doctor in the last three months with those who have not.
- 6.4 Analyse the probability of visiting the doctor. Estimate a pooled logit model, a conditional fixed effect logit model, and a random effects logit model. Make use of the lecture slides.
- 6.5 Calculate marginal effects.
- 6.6 Perform a likelihood ratio test of the random effects logit model against the pooled logit. Which conclusion do you draw?

References

- Lemieux, T., Macleod, W. B., and Parent, D. (2009). *Performance pay and wage inequality*. The Quarterly Journal of Economics, 124 (1), 1-49.
- Pfeifer, C., Stephan, G. (2019). *Why women do not ask: gender differences in fairness perceptions of own wages and subsequent wage growth*. Cambridge Journal of Economics, 43:295–310.
- Riphahn, R., Wambach, A. and Million, A. (2003). *Incentive effects in the demand for health care: a bivariate panel count data estimation*. Journal of applied econometrics, 18:397-405.